JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Finite Dimensional Vector Spaces (2005–2014)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
20	94 M	60 Min	Classic Paper Pattern (2

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	20	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q13 • ID: JAM_2014_Q13) MCQ +2 M Neg: −0.67

Q.13 Consider the subspace $W = \{(x_1, x_2, \ldots, x_{10}) \in \mathbb{R}^{10} : x_n = x_{n-1} + x_{n-2} \text{ for } 3 \le n \le 10\}$ of the vector space $\mathbb{R}^{10}$. The dimension of W is

(A) 2 (B) 3 (C) 9 (D) 10

Question 2 (JAM 2014 • Q9 • ID: JAM_2014_Q9) MCQ +1 M Neg: −0.33

Q.9 Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y,\ y + z,\ z + x)$ for all $(x, y, z) \in \mathbb{R}^3$. Then

(A) $\operatorname{rank}(T) = 0$, $\operatorname{nullity}(T) = 3$ (B) $\operatorname{rank}(T) = 2$, $\operatorname{nullity}(T) = 1$

(C) $\operatorname{rank}(T) = 1$, $\operatorname{nullity}(T) = 2$ (D) $\operatorname{rank}(T) = 3$, $\operatorname{nullity}(T) = 0$

Question 3 (JAM 2014 • Q8 • ID: JAM_2014_Q8) MCQ +1 M Neg: −0.33

Q.8 Which one of the following is a subspace of the vector space $\mathbb{R}^3$?

(A) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y = 0,\ 2x + 3z = 0\}$

(B) $\{(x, y, z) \in \mathbb{R}^3 : 2x + 3y + 4z - 3 = 0,\ z = 0\}$

(C) $\{(x, y, z) \in \mathbb{R}^3 : x \ge 0,\ y \ge 0\}$

(D) $\{(x, y, z) \in \mathbb{R}^3 : x - 1 = 0,\ y = 0\}$

Question 4 (JAM 2013 • Q10 • ID: JAM_2013_Q10) MCQ +2 M Neg: –0.67

Q.10 Let $\wp_n$ be the real vector space of all polynomials of degree at most n. Let $D : \wp_n \to \wp_{n-1}$ and $T : \wp_n \to \wp_{n+1}$ be the linear transformations defined by

$$D\left(a_0 + a_1x + a_2x^2 + \ldots + a_nx^n\right) = a_1 + 2a_2x + \ldots + na_nx^{n-1},$$
$$T\left(a_0 + a_1x + a_2x^2 + \ldots + a_nx^n\right) = a_0x + a_1x^2 + a_2x^3 + \ldots + a_nx^{n+1},$$

respectively. If A is the matrix representation of the transformation $DT - TD : \wp_n \to \wp_n$ with respect to the standard basis of $\wp_n$, then the trace of A is

(A) $-n$ (B) n (C) $n+1$ (D) $-(n+1)$

Question 5 (JAM 2013 • Q9 • ID: JAM_2013_Q9) MCQ +2 M Neg: –0.67

Q.9 Let V be the vector space of all 2×2 matrices over $\mathbb{R}$. Consider the subspaces

$$W_1 = \left\{\begin{pmatrix} a & -a \\ c & d \end{pmatrix} : a, c, d \in \mathbb{R}\right\} \text{ and } W_2 = \left\{\begin{pmatrix} a & b \\ -a & d \end{pmatrix} : a, b, d \in \mathbb{R}\right\}.$$

If $m = \dim(W_1 \cap W_2)$ and $n = \dim(W_1 + W_2)$, then the pair (m, n) is

(A) $(2, 3)$ (B) $(2, 4)$ (C) $(3, 4)$ (D) $(1, 3)$

Question 6 (JAM 2013 • Q1 • ID: JAM_2013_Q1) MCQ +2 M Neg: –0.67

Q.1 Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \\ 1 & 5 & 3 \end{pmatrix}$ and V be the vector space of all $X \in \mathbb{R}^3$ such that $AX = 0$. Then $\dim(V)$ is

(A) 0 (B) 1 (C) 2 (D) 3

Question 7 (JAM 2012 • Q11 • ID: JAM_2012_Q11) MCQ +6 M Neg: –2.0

Q.11 Consider the following subspace of $\mathbb{R}^3$:

$$W = \left\{(x, y, z) \in \mathbb{R}^3 \,\middle|\, 2x + 2y + z = 0,\ 3x + 3y - 2z = 0,\ x + y - 3z = 0\right\}.$$

The dimension of W is

(A) 0 (B) 1 (C) 2 (D) 3

Question 8 (JAM 2012 • Q10 • ID: JAM_2012_Q10) MCQ +6 M Neg: −2.0

Q.10 Let W be a vector space over $\mathbb{R}$ and let $T:\mathbb{R}^6 \to W$ be a linear transformation such that $S=\{Te_2, Te_4, Te_6\}$ spans W. Which one of the following must be TRUE ?

(A) S is a basis of W
(B) $T(\mathbb{R}^6) \neq W$
(C) $\{Te_1, Te_3, Te_5\}$ spans W
(D) $\ker(T)$ contains more than one element

MA- 2 / 36

Question 9 (JAM 2011 • Q13 • ID: JAM_2011_Q13) MCQ +6 M Neg: −2.0

Q.13 For $n \neq m$, let $T_1:\mathbb{R}^n \to \mathbb{R}^m$ and $T_2:\mathbb{R}^m \to \mathbb{R}^n$ be linear transformations such that T_1T_2 is bijective. Then

(A) rank $(T_1) = n$ and rank $(T_2) = m$
(B) rank $(T_1) = m$ and rank $(T_2) = n$
(C) rank $(T_1) = n$ and rank $(T_2) = n$
(D) rank $(T_1) = m$ and rank $(T_2) = m$

Question 10 (JAM 2011 • Q12 • ID: JAM_2011_Q12) MCQ +6 M Neg: −2.0

Q.12 Let $T:\mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation, where $n \geq 2$. For $k \leq n$, let

$$E=\{v_1, v_2, \ldots, v_k\} \subseteq \mathbb{R}^n \text{ and } F=\{Tv_1, Tv_2, \ldots, Tv_k\}.$$

Then

(A) If E is linearly independent, then F is linearly independent
(B) If F is linearly independent, then E is linearly independent
(C) If E is linearly independent, then F is linearly dependent
(D) If F is linearly independent, then E is linearly dependent

Question 11 (JAM 2010 • Q15 • ID: JAM_2010_Q15) MCQ +6 M Neg: −2.0

Q.15 Let $T:\mathbf{R}^3 \to \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, where a, b, c are real numbers not all zero. Then T

(A) is one-to-one
(B) is onto
(C) does not map any line through the origin onto itself
(D) has rank 1

Question 12 (JAM 2010 • Q14 • ID: JAM_2010_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $T:\mathbf{R}^3 \to \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis $\{e_1, e_2, e_3\}$ of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Then T

(A) maps the subspace spanned by e_1 and e_2 into itself
(B) has distinct eigenvalues
(C) has eigenvectors that span $\mathbf{R}^3$
(D) has a non-zero null space

Question 13 (JAM 2010 • Q11 • ID: JAM_2010_Q11) MCQ +6 M Neg: −2.0

Q.11 Let $T:\mathbf{R}^2 \to \mathbf{R}^2$ be a linear transformation such that $T((1,2)) = (2,3)$ and $T((0,1)) = (1,4)$. Then $T((5,6))$ is

(A) $(6,-1)$ (B) $(-6,1)$ (C) $(-1,6)$ (D) $(1,-6)$

Question 14 (JAM 2009 • Q4 • ID: JAM_2009_Q4) MCQ +6 M Neg: −2.0

Q.4 Let $T: \mathrm{R}^4 \to \mathrm{R}^4$ be a linear transformation satisfying $T^3 + 3T^2 = 4I$, where I is the identity transformation. Then the linear transformation $S = T^4 + 3T^3 - 4I$ is

(A) one-one but not onto
(B) onto but not one-one
(C) invertible
(D) non-invertible

Question 15 (JAM 2009 • Q1 • ID: JAM_2009_Q1) MCQ +6 M Neg: −2.0

Q.1 Let V be the vector space of all 6×6 real matrices over the field $\mathbb{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is

(A) 15 (B) 18 (C) 21 (D) 35

Question 16 (JAM 2008 • Q11 • ID: JAM_2008_Q11) MCQ +6 M Neg: −2.0

Q.11 Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that $T(1,2,3) = (1,2,3)$, $T(1,5,0) = (2,10,0)$ and $T(-1,2,-1) = (-3,6,-3)$. The dimension of the vector space spanned by all the eigenvectors of T is

(A) 0 (B) 1 (C) 2 (D) 3

Question 17 (JAM 2008 • Q2 • ID: JAM_2008_Q2) MCQ +6 M Neg: −2.0

Q.2 Let $S = \{T : \mathbb{R}^3 \to \mathbb{R}^3 \mid T$ is a linear transformation with $T(1,0,1) = (1,2,3)$, $T(1,2,3) = (1,0,1)\}$. Then S is

(A) a singleton set (B) a finite set containing more than one element
(C) a countably infinite set (D) an uncountable set

Question 18 (JAM 2007 • Q5 • ID: JAM_2007_Q5) MCQ +6 M Neg: −2.0

5. Which of the following sets is a basis for the subspace

$$W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t = 0,\ y + t = 0 \right\}$$

of the vector space of all real 2×2 matrices?

(A) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

(B) $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

(C) $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$

(D) $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

Question 19 (JAM 2006 • Q8 • ID: JAM_2006_Q8) MCQ +6 M Neg: −2.0

8. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2,\ x_1 - x_2,\ 0).$$

If $N(T)$ and $R(T)$ denote the null space and the range space of T respectively, then

(A) $\dim N(T) = 2$

(B) $\dim R(T) = 2$

(C) $R(T) = N(T)$

(D) $N(T) \subset R(T)$

Question 20 (JAM 2005 • Q7 • ID: JAM_2005_Q7) MCQ +6 M Neg: −2.0

7. Let $T : \boldsymbol{R}^3 \longrightarrow \boldsymbol{R}^3$ be a linear transformation and I be the identity transformation of $\boldsymbol{R}^3$. If there is a scalar c and a non-zero vector $x \in \boldsymbol{R}^3$ such that $T(x) = cx$, then rank$(T - cI)$

(A) cannot be 0

(B) cannot be 1

(C) cannot be 2

(D) cannot be 3

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	**A**
2	MCQ	+1	**D**
3	MCQ	+1	**A**
4	MCQ	+2	**C**
5	MCQ	+2	**B**
6	MCQ	+2	**B**
7	MCQ	+6	**B**
8	MCQ	+6	**D**
9	MCQ	+6	**D**
10	MCQ	+6	**B**

Q. No.	Type	Marks	Official Key
11	MCQ	+6	**C**
12	MCQ	+6	**C**
13	MCQ	+6	**A**
14	MCQ	+6	**D**
15	MCQ	+6	**C**
16	MCQ	+6	**D**
17	MCQ	+6	**D**
18	MCQ	+6	**D**
19	MCQ	+6	**A**
20	MCQ	+6	**D**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.